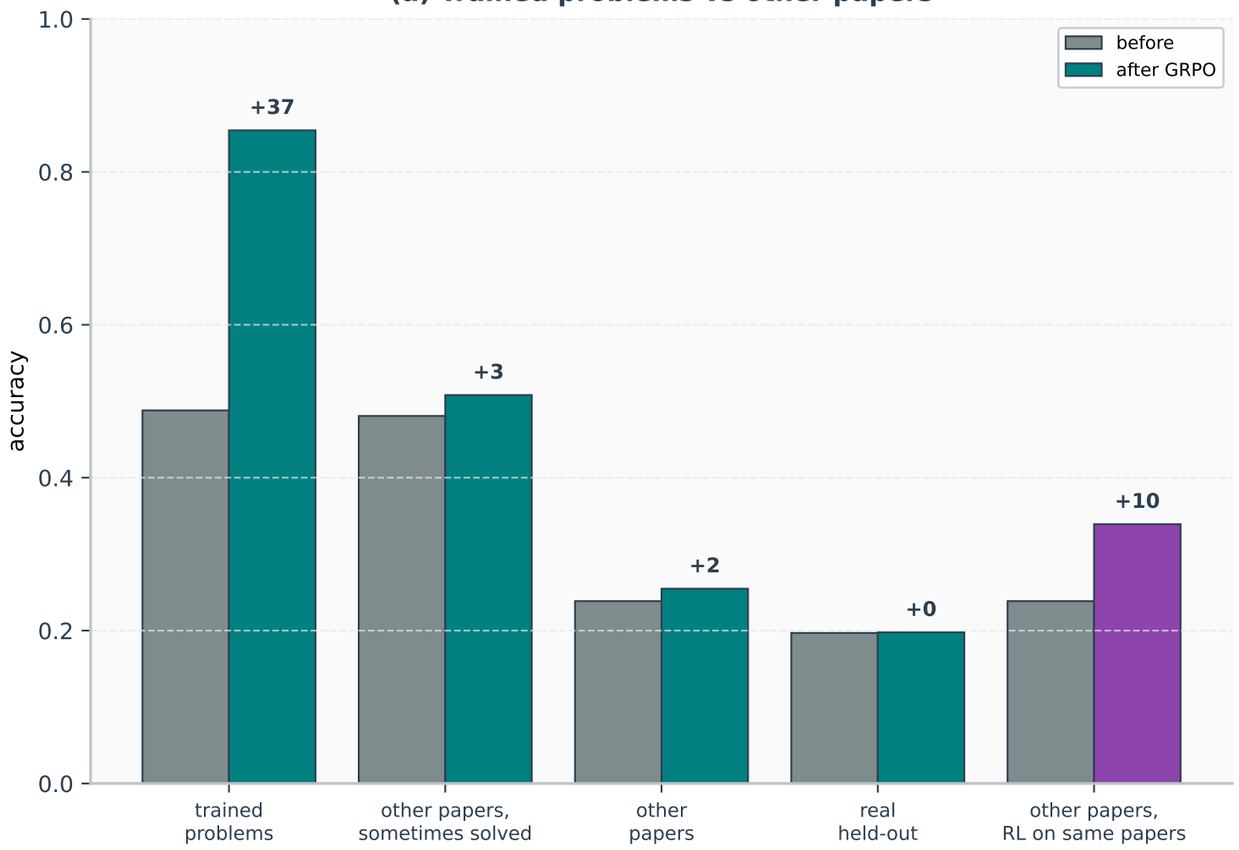
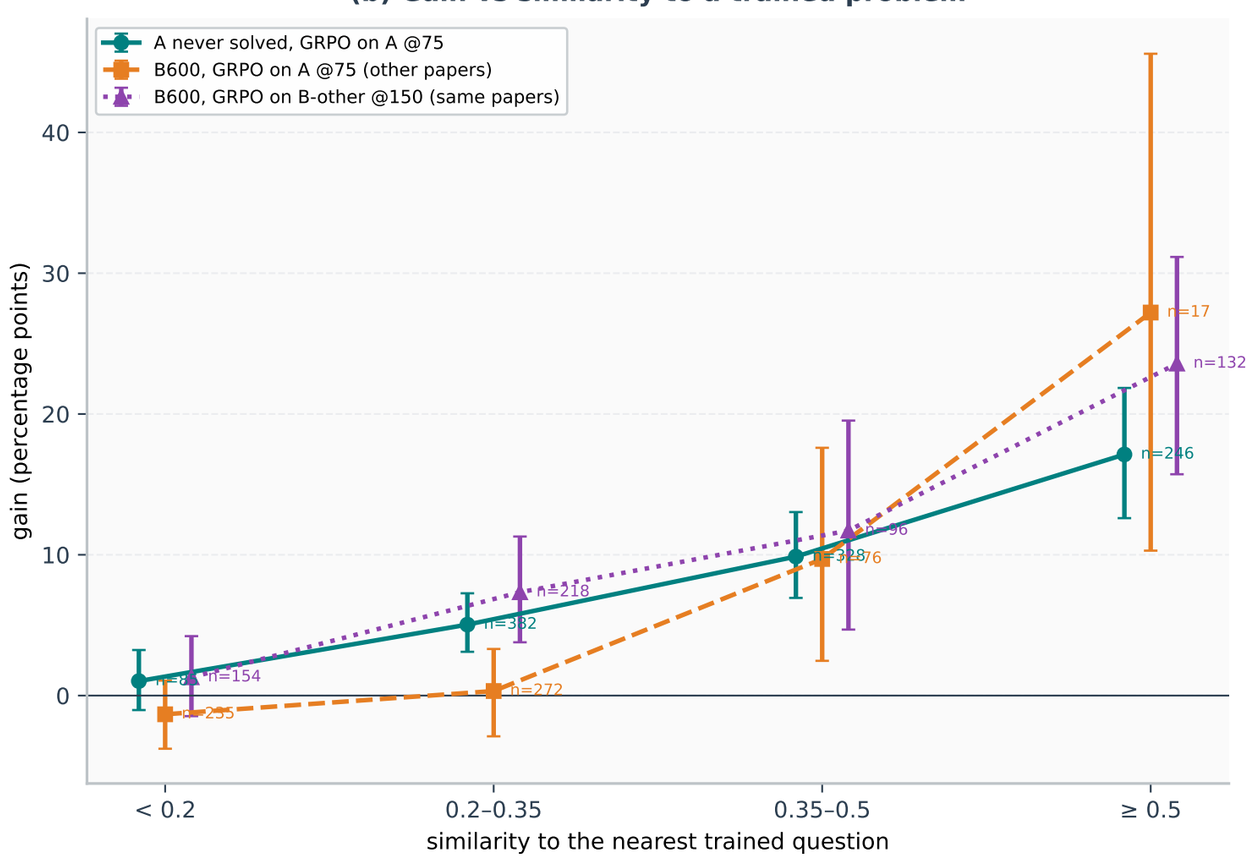
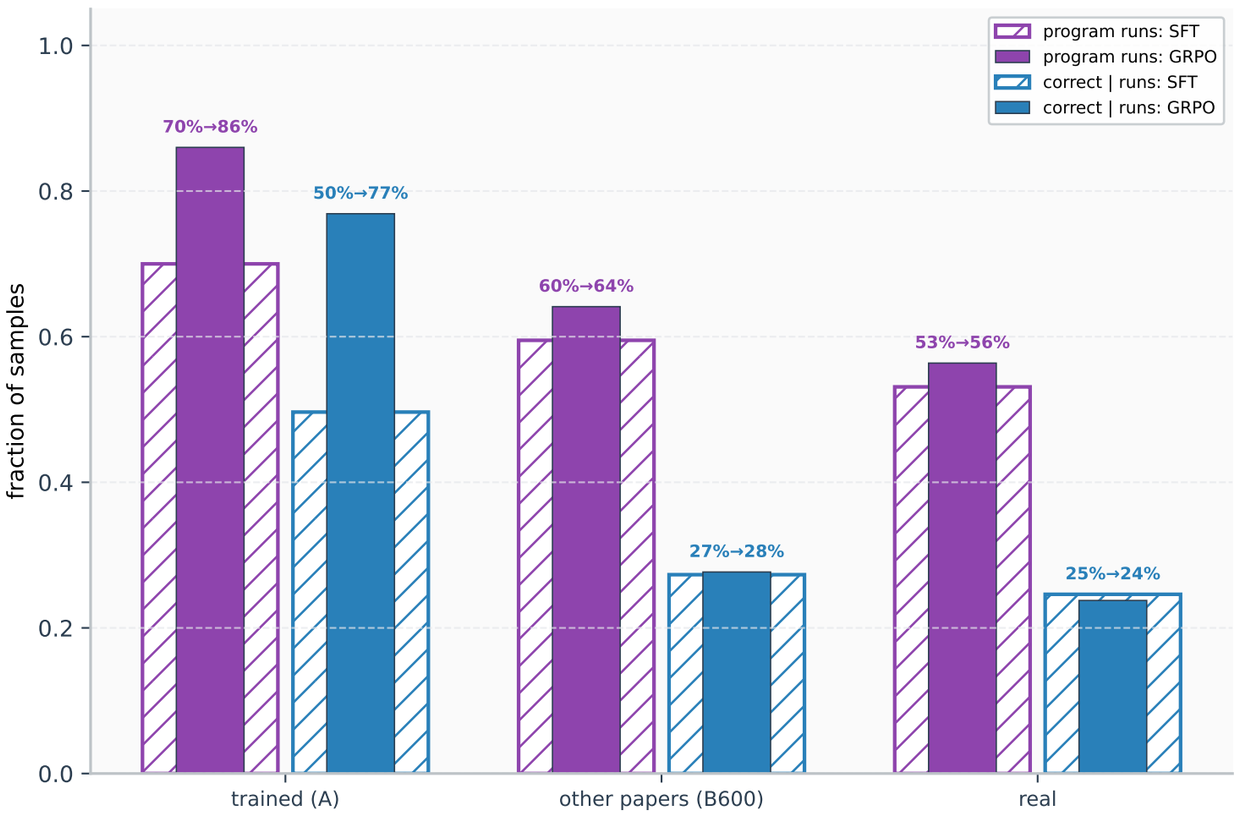

(a) Trained problems vs other papers
before
after GRPO
accuracy
1.0
0.8
0.6
0.4
0.2
0.0
+37
+3
+2
+0
+10
trained problems
other papers, sometimes solved
other papers
real held-out
other papers, RL on same papers


(b) Gain vs similarity to a trained problem
A never solved, GRPO on A @75
B600, GRPO on A @75 (other papers)
B600, GRPO on B-other @150 (same papers)
gain (percentage points)
40
30
20
10
0
n=154
n=218
n=382
n=272
n=255
n=328
n=376
n=96
n=17
n=132
n=246
< 0.2
0.2–0.35
0.35–0.5
≥ 0.5
similarity to the nearest trained question


(c) Valid program vs right setup
program runs: SFT
program runs: GRPO
correct | runs: SFT
correct | runs: GRPO
fraction of samples
1.0
0.8
0.6
0.4
0.2
0.0
70%→86%
50%→77%
60%→64%
27%→28%
53%→56%
25%→24%
trained (A)
other papers (B600)
real

(d) How the policy changed
change in log-probability per token
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
−0.150
KL 0.06
KL 0.06
KL 0.06
SFT samples: correct
SFT samples: runs, wrong value
SFT samples: crashes
trained (A)
other papers (B600)
real